XMBA 201A – Managerial Economics

Office Hours Practice Problems: Block 2 Review

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Question 1: Marginal Revenue, Elasticity, and the Lerner Rule

Fall 2021 Final Exam

The Cal Student Store is the monopoly seller of children's sized Oski-the-bear Halloween costumes. They incur both fixed and variable costs selling the costumes. Because they are worried about re-sales on e-Bay, they decided not to engage in any form of price discrimination. That is, they find it necessary to charge the same price per costume. Having conducted a market research survey last week, the Cal Student Store believes that the quantity of outfits demanded this season is given by the following formula: $q = 1{,}600 - 10p$ (q is the number of costumes sold and p is the price per costume). The Student Store decided to charge a price of $30 per costume, at which management knows that they're making positive profits, but does not know exactly what it costs to sell each outfit. Is this price too high, too low, or do you need more information to answer the question?

Question 2: Group Price Discrimination and Demand Aggregation

Fall 2021 Final Exam

Demand for entrance to the theme park Six Flags Marine World can be divided into three customer segments: young kids under the age of 6, older kids ages 6 to 18, and adults. Each segment's demand is given as follows:

Young kids: $qY = 600 - 50\,pY$
Older kids: $qO = 2{,}000 - 100\,pO$
Adults: $qA = 2{,}400 - 80\,pA$

where qi measures the quantity demanded (that enters the park) for group i, and pi is the entrance price in dollars for group i. Assume that Six Flags' marginal cost per theme park visitor is constant and equal to $10 for any visitor regardless of age. In other words, kids put no more wear and tear on the rides and create no more messes to clean up than adults. Also assume that the park is huge and there are no space limitations.

a. [15] A consultant convinces Six Flags Marine World that they can perfectly distinguish kids under 6 from other kids by using the kid's height and by relying on the parents to accurately represent their kid's age. Assuming they can charge separate prices to each of the three customer segment groups, what would those prices be? What would profits be and how many customers from each group would visit the park?

b. [15] A local politician thinks that it is unfair to charge young and older kids different prices and decides to start a petition among local families asking Six Flags Marine World to charge only two different prices: pA for adults and pK for all kids. From a purely economic standpoint (e.g., putting aside considerations about whether kids should go to theme parks) do you think he would get support? If so, who would support the petition? If not, why not?

Question 3: Two-Part Pricing and Screening Heterogeneous Consumers

Fall 2021 Final Exam

A ski resort has a potential market of 10,000 skiers. Of those, 5,000 are avid skiers with a willingness to pay of $100 per ski day and 5,000 are casual skiers with a willingness to pay of $60 per ski day. The resort cannot directly price-discriminate between the two types of skiers: every type of ticket it sells can be bought by any type of skier. The marginal cost of an extra skier on the slopes is zero.

a. [5] How much should the resort charge for a ski pass to maximize profit?

Now suppose the resort is considering building a mountain-top spa that can be used by skiers between runs. The marginal cost of an extra spa user is zero. The resort would then be able to sell two types of tickets: a basic ski pass and a ski pass that includes spa access. The willingness to pay for a basic ticket is the same as above. The willingness to pay for a ticket that includes both skiing and spa access is $150 for an avid skier and $80 for a casual one.

b. [15] What is the most the resort would be willing to pay to build the spa? What prices would it charge for the different tickets it sells?

P1

$q(p) = 1600 - 10p$ ✓✓

$q(30) = 1600 - 10 \times 30$

$= 1600 - 300$

$= 1300$

$\frac{dq}{dp} = -10$

$\varepsilon_d = \frac{p}{q} \cdot \frac{dq}{dp} = \frac{30}{1300} \times -10 = \frac{-300}{1300} = -0.23$

$|\varepsilon_d| = 0.23 < 1$ ✓

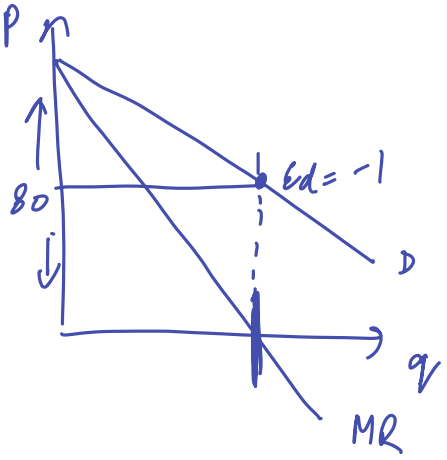
$MR < 0$ ✓

$MR = p\left(1 + \frac{1}{\varepsilon_d}\right)$ LERNER'S MARKUP.

$= 30\left(1 + \frac{1}{-0.23}\right)$

$= -100$

$\varepsilon_d = -1$



$-1 = -10 \cdot \frac{p}{q}$

$q = 10p$

$1600 - 10p = 10p$

$1600 = 20p$

$80 = p$

Q2.

Segment	Inverse demand	MR	Set MR = MC = 10
Young Kids	$12 - q/50$	$12 - q/25$	$12 - q/25 = 10$
Old Kids	$20 - q/100$	$20 - q/50$	$20 - q/50 = 10$
Adults	$30 - q/80$	$30 - q/40$	$30 - q/40 = 10$

q^*	p^*
50	11
500	15
800	20

$q = 600 - 50p$

$50p = 600 - q$

$p = \frac{600}{50} - \frac{q}{50}$

$= 12 - q/50$

$q = 600 - 50p \rightarrow 600p - 50p^2$

$TR = p \cdot q \Rightarrow$

$MR = \frac{\partial TR}{\partial q}$

when $p = a - bq$ then $MR = a - 2bq$

$MR = MC$

$12 - \frac{q}{25} = 10$

$\Rightarrow 12 \times 25 - q = 250$

$\Rightarrow 12 \times 25 - 10 \times 25 = q$

$\Rightarrow (12 - 10)25 = q$

$\Rightarrow 2 \times 25 = q$

$\Rightarrow 50 = q^*$

$$\pi = \sum_{i=1}^{3} (p_i - MC) \cdot q_i$$

$$= (11-10)50 + (15-10)500 + (20-10)800$$

$$= 50 + 2500 + 8000$$

$$= 10,550.$$

b) $Q_K(p) = q_Y(p) + q_O(p)$

$q_Y = 600 - 50p_Y$ ✓

$q_O = 2000 - 100p_O$

$p_Y = 12 - q/50$ ✓

$p_O = 20 - q/100$ ✓

For $20 \geq p > 12$ only have older kids demanding tickets.

$Q_K = 2000 - 100p_O$ $[0 - 800]$ q

For $12 \leq p < 0$ young + old kids.

$q_K = 2600 - 150 P_K$

If $p_K = 0$ then $q_K = 2600$

If $p_K = 12$ then $q_K = 2600 - 12 \times 150$

$= 2600 - 1800$

$= 800.$

Region 1: $q^* = 500$ ✓ $p^* = 15.$

Region 2: $\| q_K = 2600 - 150 P_K$

$150 P_K = 2600 - q_K$

$P_K = \frac{2600}{150} - \frac{q_K}{150}$

$= 17.33 - q_K/150$ Inverse demand

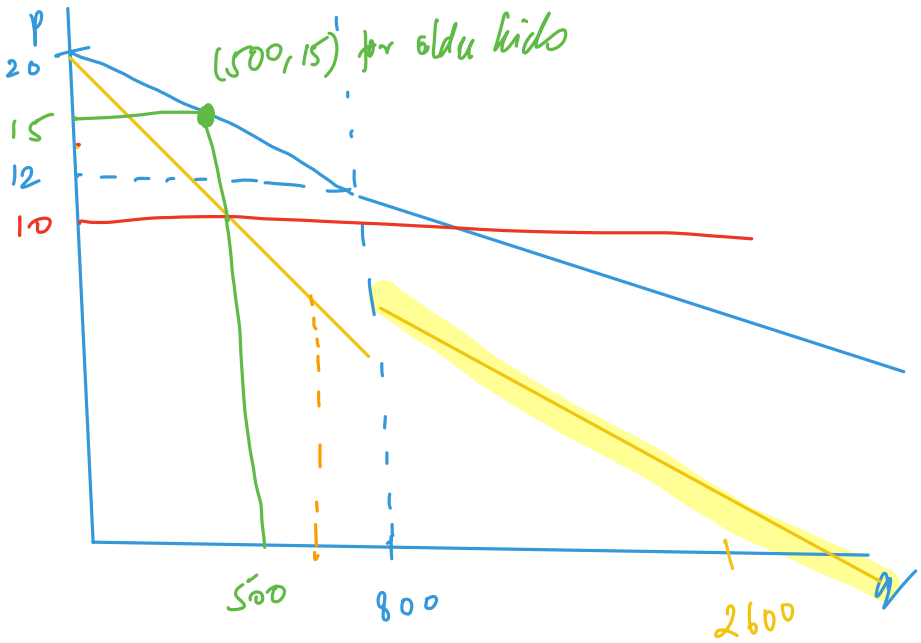
$p = a - bq$

$\Rightarrow MR = a - 2bq$

$MR = 17.33 - \frac{q_k}{75}$

$10 = 17.33 - \frac{q_k}{75} \Rightarrow q_k^* = 550$

800 − 2600 !



Q3 a

Revenues.

$100 → 5000 × 100 = 500,000 £

$60 → (5000 + 5000) 60 = 600,000 £

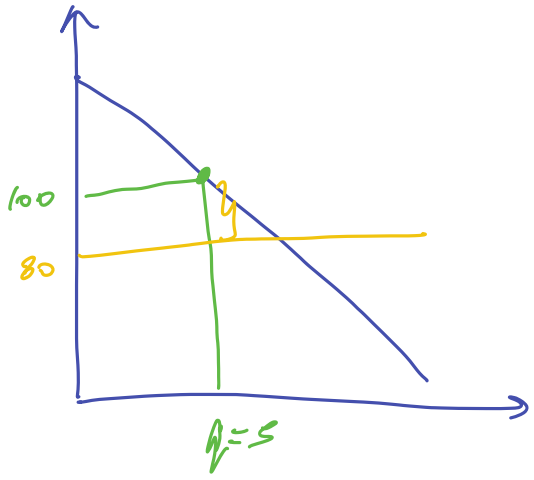
	Basic Ski NTP	Ski + Spa WTP
Avid	$100	$150
Casual	$60	$80

Basic Ski → $60

Price Ski + Spa s.t. $150 - p_{combo} \geq 40$.

$150 - p_c = 40$

$p^*_{combo} = 110$



$\pi = 5000 \times 110 + 5000 \times 60$

$= 550000 + 300000$

$= 850,000$